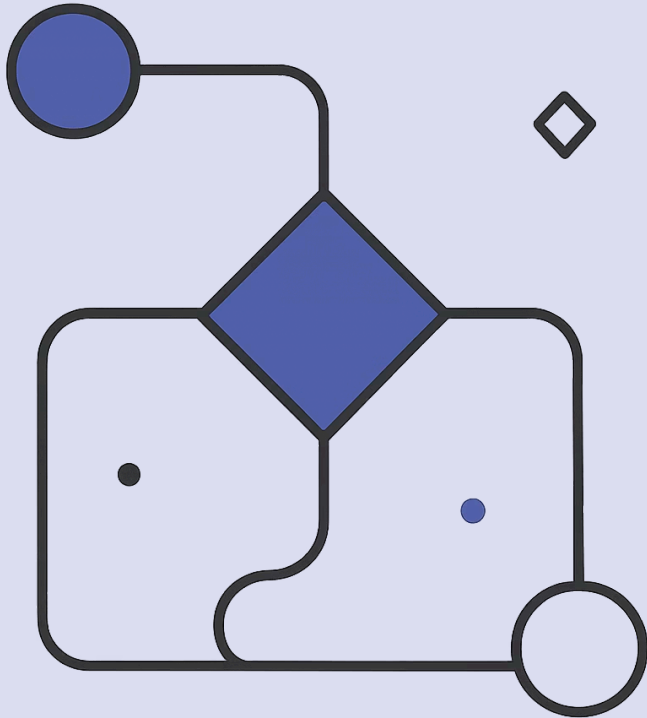




Logistic Regression

First classification model for predicting probabilities and classes.

What Students Will Learn

01

Classification Foundations

Why Logistic Regression is used for classification, probability, sigmoid, and threshold

02

The Formula

Logistic Regression formula, coefficient interpretation, and log loss

03

Evaluation

Confusion matrix, metrics, threshold tuning, practical mistakes and limitations

Classification Predicts Categories

Student Result

Pass / Fail

Email

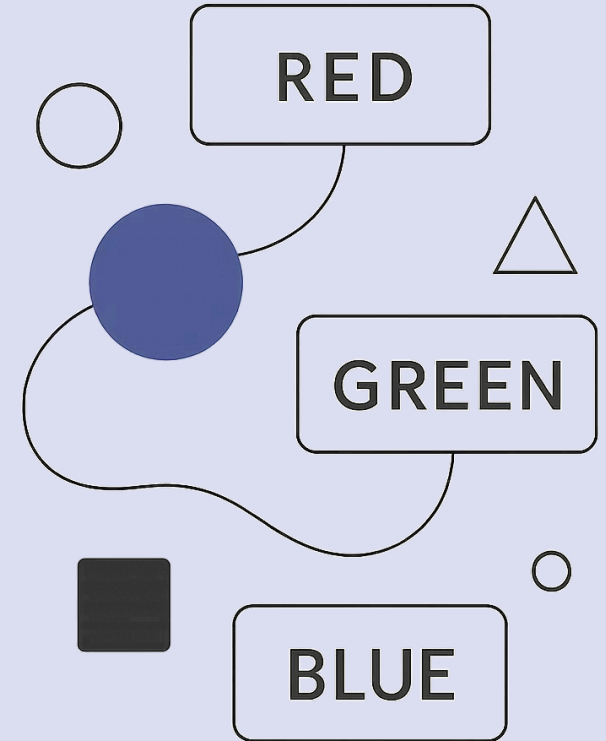
Spam / Not Spam

Loan

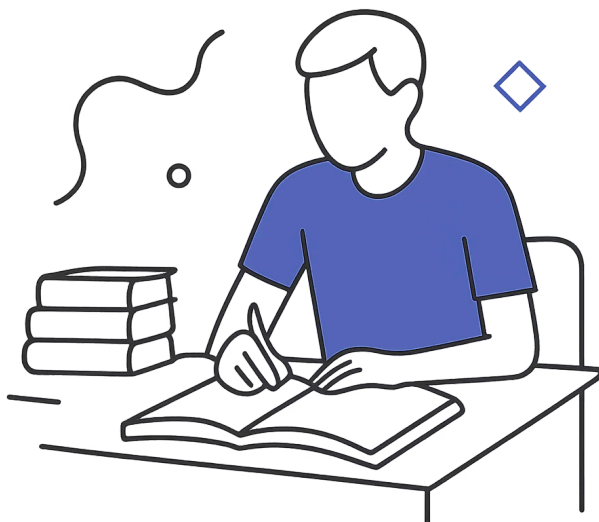
Approved / Rejected

Patient

Disease / No Disease



Example: Student Pass Prediction



Question

Can we predict whether a student will pass?

Feature	Example
Study Hours	6
Attendance	80%
Previous Score	60

 Target: Pass / Fail

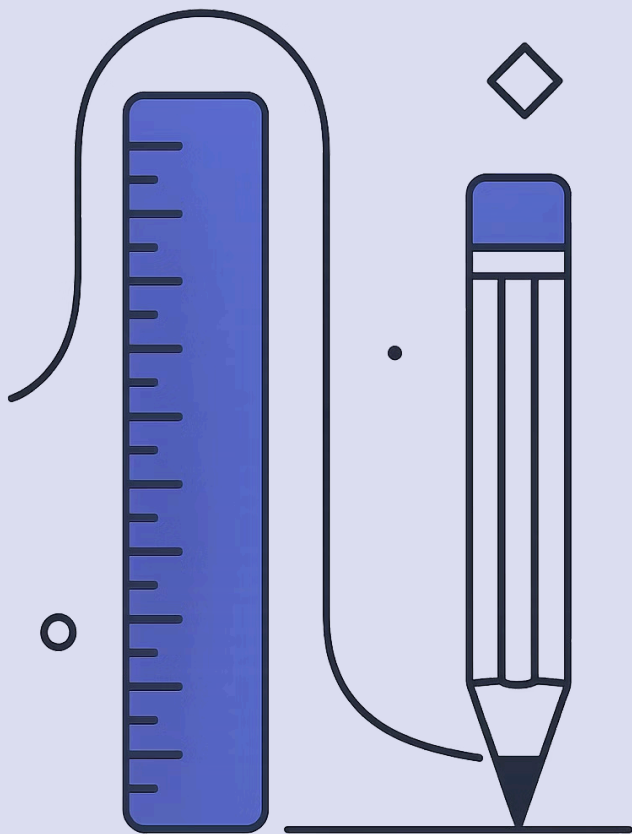
Example Dataset

Study Hours	Attendance	Previous Score	Result
2	55	40	Fail
4	70	55	Pass
3	65	45	Fail
6	85	70	Pass
7	90	80	Pass

Features vs Target

Column	Role
Study Hours	Feature
Attendance	Feature
Previous Score	Feature
Result	Target

- ❏ The target column is categorical — it contains class labels, not continuous numbers.



Why Not Linear Regression?

Linear Regression predicts continuous numbers — but classification needs a probability between 0 and 1.

⚠ Linear Regression can output invalid probabilities like -0.3 , 1.4 , or 2.1 .

Logistic Regression Output



Predicts Probability

Example: Probability of Pass = 0.82

Then Makes a Decision

If probability $\geq$ threshold $\rightarrow$ Pass

If probability $<$ threshold $\rightarrow$ Fail

The Threshold

Default threshold is usually 0.50.

Probability of Pass	Prediction
0.82	Pass
0.65	Pass
0.40	Fail
0.20	Fail

Step 1: Linear Score

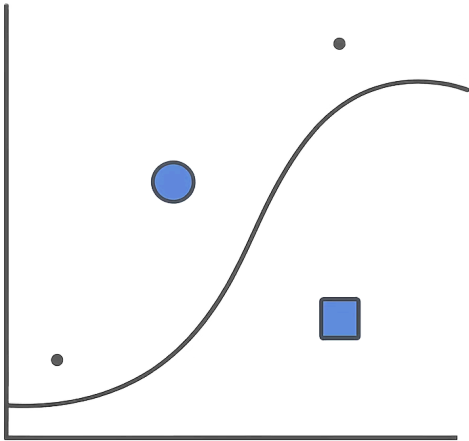
Logistic Regression first calculates a linear score z :

$$z = b_0 + b_1x_1 + b_2x_2 + b_3x_3$$

For students:

$$z = b_0 + b_1(\text{Study Hours}) + b_2(\text{Attendance}) + b_3(\text{Previous Score})$$

Step 2: Sigmoid Function



The linear score z is converted into a probability using the sigmoid function:

$$p = 1 / (1 + e^{-z})$$

- ✓ Output is always between 0 and 1.

Sigmoid Intuition

z value	Probability
-3	0.05
-1	0.27
0	0.50
1	0.73
3	0.95

 Higher z $\rightarrow$ higher probability.

The Complete Formula

1

Linear Score

$$z = b_0 + b_1x_1 + b_2x_2 + b_3x_3$$

2

Probability

$$p = 1 / (1 + e^{-z})$$

3

Prediction

If $p \geq 0.50 \rightarrow$ Class 1 | If $p < 0.50 \rightarrow$ Class 0

Class Encoding

For binary classification, classes are encoded as numbers:

Class	Encoded Value
Fail	0
Pass	1

 p = probability of class 1. Example: p = Probability of Pass.

Prediction Example

Suppose the model learned:

$$z = -8 + 0.7(\text{Study Hours}) + 0.05(\text{Attendance}) + 0.08(\text{Previous Score})$$

Feature	Value
Study Hours	6
Attendance	80
Previous Score	60

Calculating z

1 Substitute values

$$z = -8 + 0.7(6) + 0.05(80) + 0.08(60)$$

2 Expand

$$z = -8 + 4.2 + 4.0 + 4.8$$

3 Result

$z = 5.0 \rightarrow$ now convert to probability

Convert z to Probability

$$p = 1 / (1 + e^{-5}) \approx 0.99$$

Probability of Pass

$\approx 99\%$

Prediction

Pass ✓

Contribution Breakdown

Part	Calculation	Contribution
Intercept	-8	-8.0
Study Hours	0.7×6	4.2
Attendance	0.05×80	4.0
Previous Score	0.08×60	4.8
Linear Score z		5.0

Coefficient Meaning

Positive Coefficient

Feature increases probability of class 1

Negative Coefficient

Feature decreases probability of class 1

Near Zero

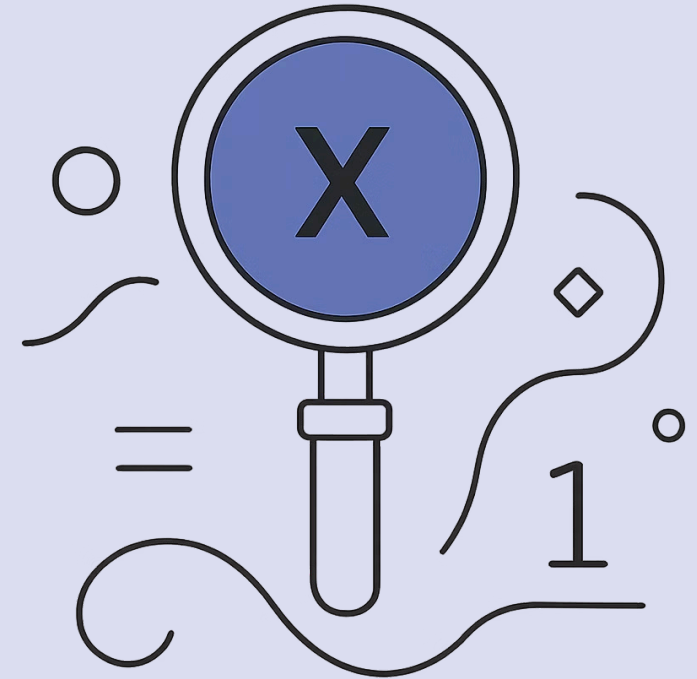
Feature has a weak effect

 A positive study hours coefficient means more study increases pass probability.

Interpreting Coefficients

Effect of one feature while all other features stay fixed.

If the Study Hours coefficient is positive: more study hours increase pass probability, *keeping attendance and previous score fixed*.



Odds



Formula

$$\text{Odds} = p / (1 - p)$$

Example

If $p = 0.80$:

$$\text{Odds} = 0.80 / 0.20 = 4$$

i Pass is 4 times as likely as Fail.

“Probability compares success to total. Odds compare success to failure.”

Log-Odds

$$\text{log-odds} = b_0 + b_1x_1 + b_2x_2 + \dots$$

$$\log(p / (1 - p)) = b_0 + b_1x_1 + b_2x_2 + \dots$$

$$z = \text{log-odds}$$

What it means

Logistic Regression does not model probability linearly — it models log-odds linearly.

Odds Ratio

Odds Ratio = e^b

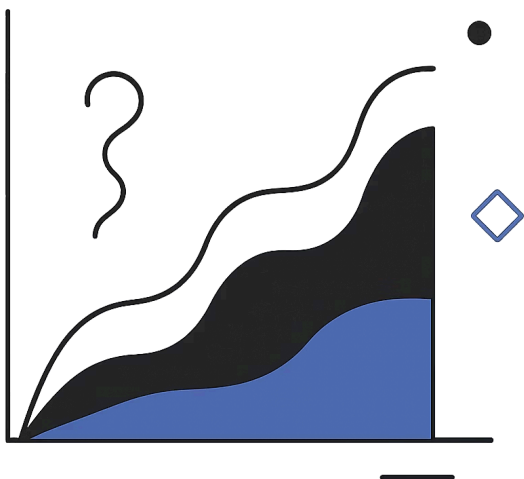
If coefficient $b = 0.7$:

Odds Ratio = $e^{0.7} \approx 2.01$

 For 1 unit increase in that feature, odds become about 2×, keeping other features fixed.

Decision Boundary

Logistic Regression separates classes using a boundary. At threshold 0.50, $\rho = 0.50$, which occurs when $z = 0$.



$$b_0 + b_1x_1 + b_2x_2 = 0$$

z value	Probability	Prediction
$z > 0$	$\rho > 0.50$	Class 1
$z = 0$	$\rho = 0.50$	Boundary
$z < 0$	$\rho < 0.50$	Class 0

This is the decision boundary.

Linear Decision Boundary

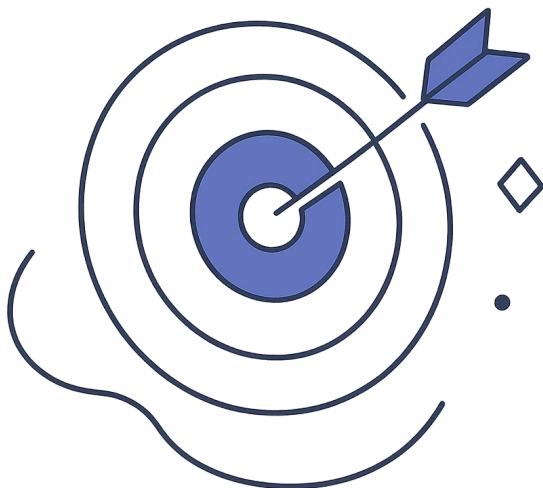
Feature Space	Boundary
2 features	Line
3 features	Plane
Many features	Hyperplane

Logistic Regression always creates a linear decision boundary.

When Logistic Regression Works Well

- Classes are mostly linearly separable
- Interpretability is important
- Probability output is needed
- Dataset is tabular
- You need a strong baseline classifier

Training Goal



The model learns coefficients: $b_0, b_1, b_2, b_3 \dots$

Actual Pass

Predicted probability should be high

Actual Fail

Predicted probability should be low

Actual Class	Good Prediction	Bad Prediction
Pass = 1	$\rho = 0.95$	$\rho = 0.10$
Fail = 0	$\rho = 0.05$	$\rho = 0.90$

Why Not MSE?

Linear Regression

Uses MSE — minimises squared error on continuous values

Logistic Regression

Uses Log Loss — classification is probability-based

 Log Loss strongly penalises confident wrong predictions.

Log Loss Formula

For one record, where p = predicted probability of class 1:

If actual $y = 1$

Loss = $-\log(p)$

If actual $y = 0$

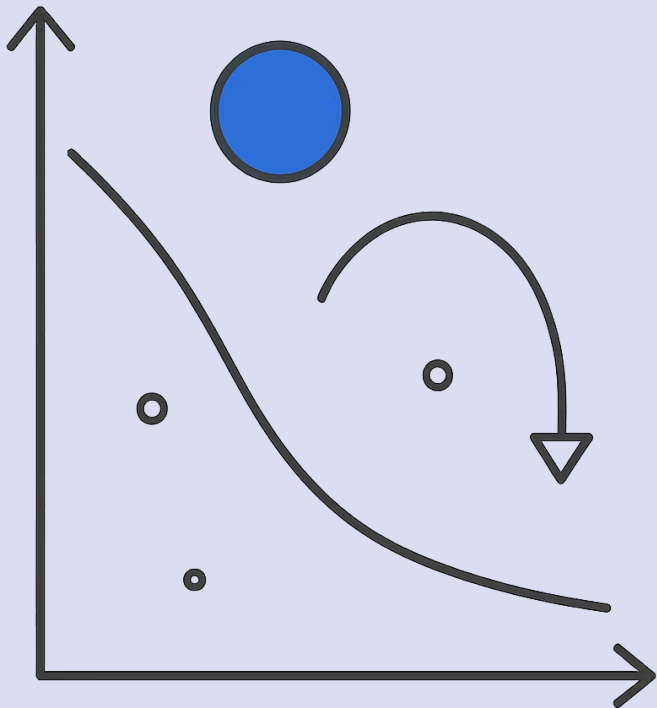
Loss = $-\log(1 - p)$

Log Loss Intuition

Actual Class	Predicted Probability	Result
1	0.99	Very small loss
1	0.60	Medium loss
1	0.10	Very high loss
0	0.05	Very small loss
0	0.90	Very high loss



Confident wrong predictions are punished heavily.



Training Objective

Find coefficients that produce good probabilities — minimise Log Loss.

Good Model

High probability for correct class, low probability for wrong class

Regularisation

Logistic Regression can overfit when features are many. Regularisation controls coefficient size.

L2

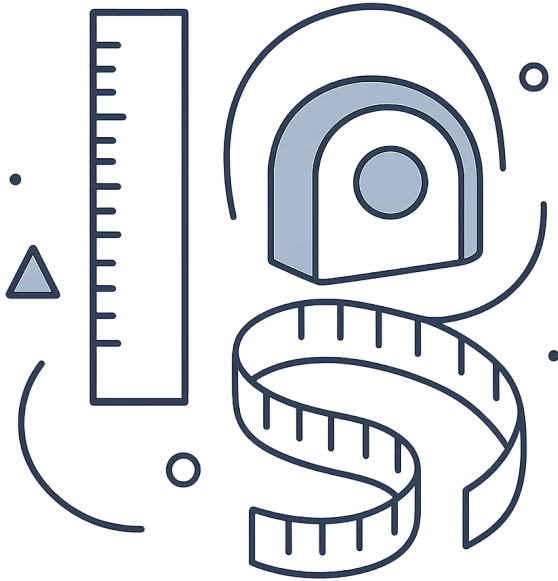
Shrinks all coefficients

L1

Can make weak coefficients zero

✓ Both methods help generalisation.

Feature Scaling



Why Scale?

- Helps optimisation
- Makes regularisation fair
- Makes coefficients easier to compare

Common Method

$$z = (x - \text{mean}) / \text{standard deviation}$$

Evaluation Starts Here

After training, we evaluate predictions. Classification evaluation differs from regression.



Confusion Matrix

Shows types of mistakes



Accuracy, Precision, Recall, F1

Core performance metrics



ROC-AUC

Measures class separation

Confusion Matrix

	Predicted Pass	Predicted Fail
Actual Pass	True Positive	False Negative
Actual Fail	False Positive	True Negative

This table shows what type of mistakes the model makes.

Mistake	Meaning
False Positive	Healthy person predicted as diseased
False Negative	Diseased person predicted as healthy

Confusion Matrix Example

	Predicted Pass	Predicted Fail
Actual Pass	80	20
Actual Fail	10	90

TP

80

FN

20

FP

10

TN

90

Accuracy

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Measures the proportion of correct predictions out of all predictions.

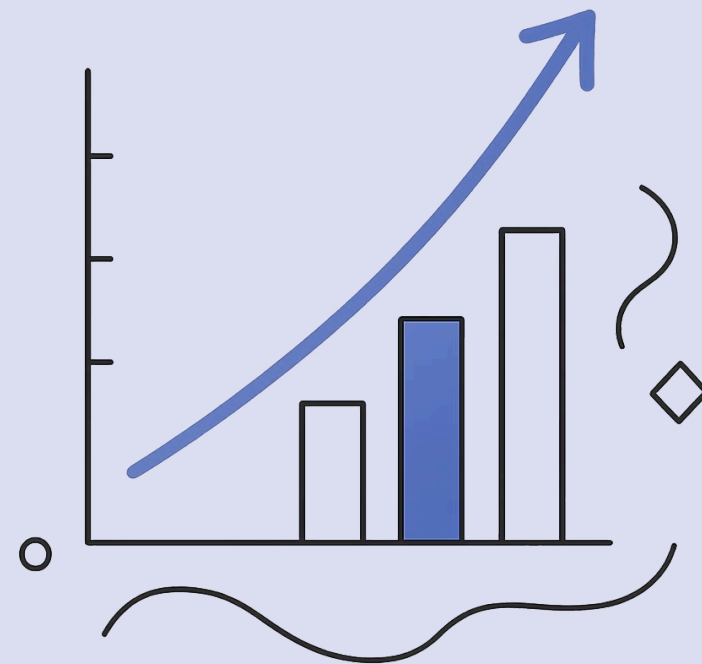
- ⓘ Works well when classes are balanced.

The Accuracy Problem

Suppose 99% of students pass. A model that predicts everyone passes achieves 99% accuracy — but cannot identify failing students.

⚠ Accuracy can be misleading when classes are imbalanced.

Class	Count
Pass	990
Fail	10



Precision

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Out of predicted positives, how many were truly positive?

Use When

False positives are costly

Example

Do not wrongly mark a good email as spam

Recall

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Out of actual positives, how many did we catch?

Use When

False negatives are costly

Example

Do not miss disease cases

Precision vs Recall

Metric	Focus	Costly Mistake
Precision	Positive predictions should be correct	False Positive
Recall	Actual positives should be caught	False Negative

 Choose based on the business problem.

F1 Score

$$F1 = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$$

Balances precision and recall into one score. Use when:

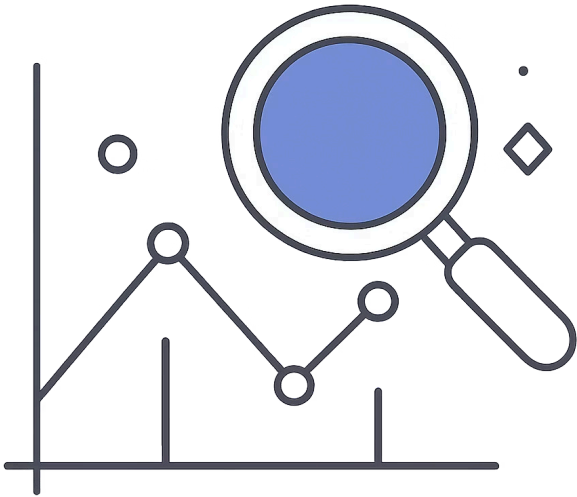
- Dataset is imbalanced
- Both precision and recall matter
- You need one combined score

ROC-AUC

Measures class separation across all thresholds. Useful for comparing classifiers.

AUC	Meaning
0.5	Random guessing
0.7	Fair
0.8	Good
0.9+	Strong

Precision-Recall AUC



When to Use PR-AUC

When the positive class is rare:

- Fraud detection
- Disease detection
- Defect detection

i With highly imbalanced data, PR-AUC may be more informative than ROC-AUC.

Threshold Tuning

Default Threshold

0.50 — not always best

Example: Disease Detection

Use threshold = 0.30 to catch more cases, accepting more false positives

Threshold Trade-Off

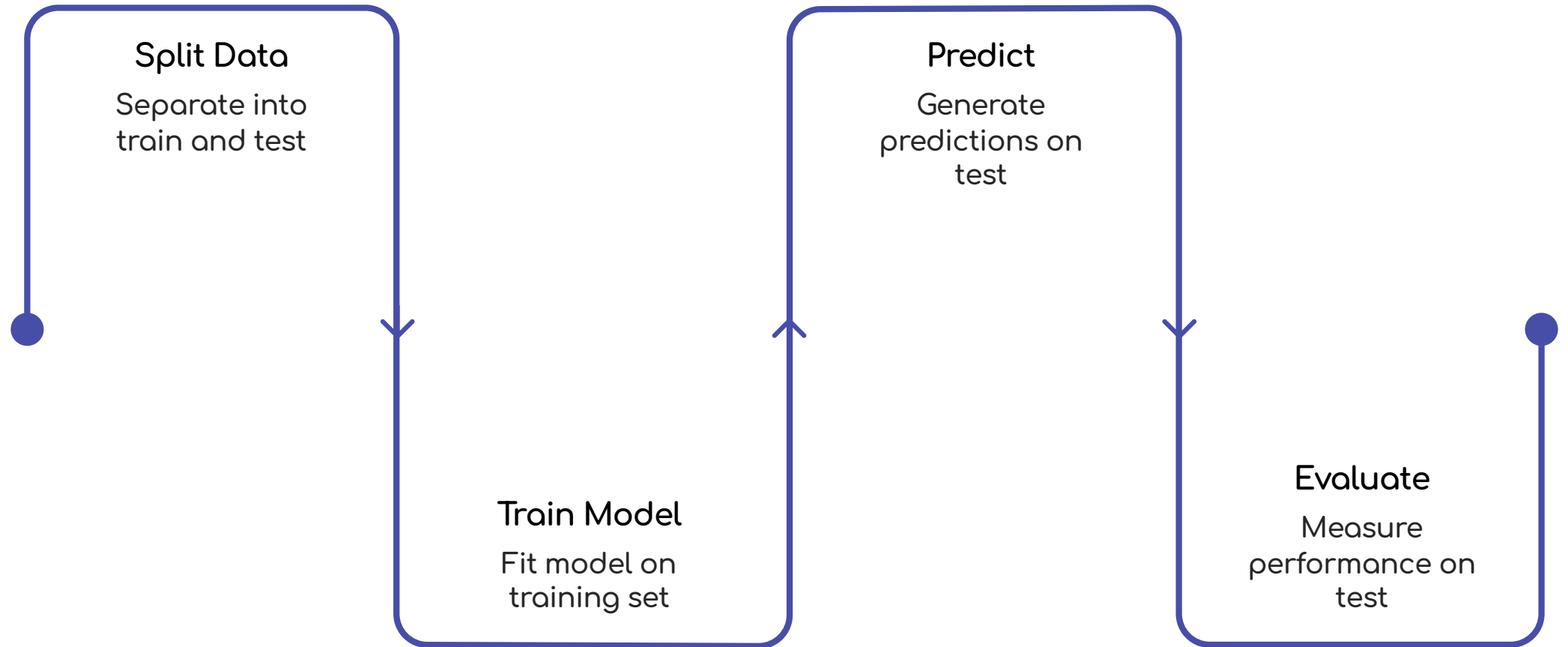
Lower Threshold	Higher Threshold
More positives predicted	Fewer positives predicted
Higher recall	Higher precision
More false positives	More false negatives

❏ Threshold should match business cost.

Business Threshold Examples

Problem	Priority	Strategy
Disease detection	Catch positives	Lower threshold
Spam detection	Avoid false spam	Higher precision
Fraud detection	Catch fraud	High recall
Loan approval	Avoid risky approvals	Balance precision & recall

Train-Test Split



⚠ Test data must remain unseen during training.

Imbalanced Classes

Class	Count
Pass	950
Fail	50

The Problem

The model may ignore the minority class. Accuracy looks high, but recall for failing students may be very poor.

In imbalanced data, high accuracy can hide model failure.”

Problem	Majority Class	Minority Class
Fraud detection	Not Fraud	Fraud
Disease detection	No Disease	Disease
Student risk prediction	Safe	At Risk
Machine failure	Normal	Failure

Handling Imbalance



Class Weights

Penalise minority mistakes more



Oversampling / SMOTE

Add or synthesise minority samples



Undersampling

Reduce majority samples



Threshold Tuning

Adjust the decision threshold

Class	Count
Pass	950
Fail	950

Categorical Features

Logistic Regression needs numeric input — text categories must be encoded.

Feature	Encoding
Gender	One-hot encoding
City	One-hot encoding
Department	One-hot encoding
Skill Level	Ordinal encoding (if ordered)

One-Hot Encoding Example

Original

School Type
Government
Private

Encoded

School Type_Private
0
1



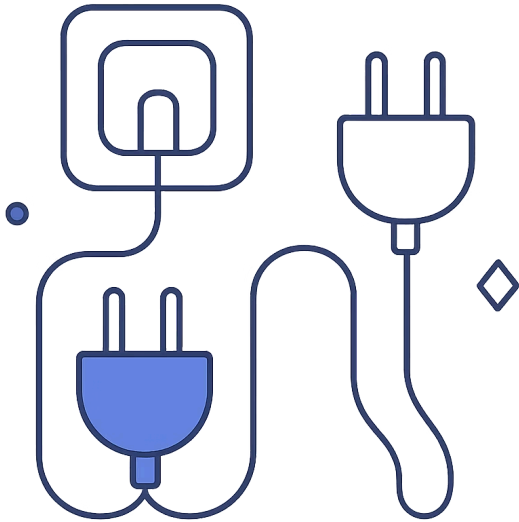
Government = reference category (0).

Data Leakage

Target leakage means a feature secretly contains the answer, creating fake high performance.

Feature	Problem
Final result status	Already contains target
Certificate issued	Decided after result
Rank after exam	Future information
Pass notification sent	Depends on target

Multicollinearity

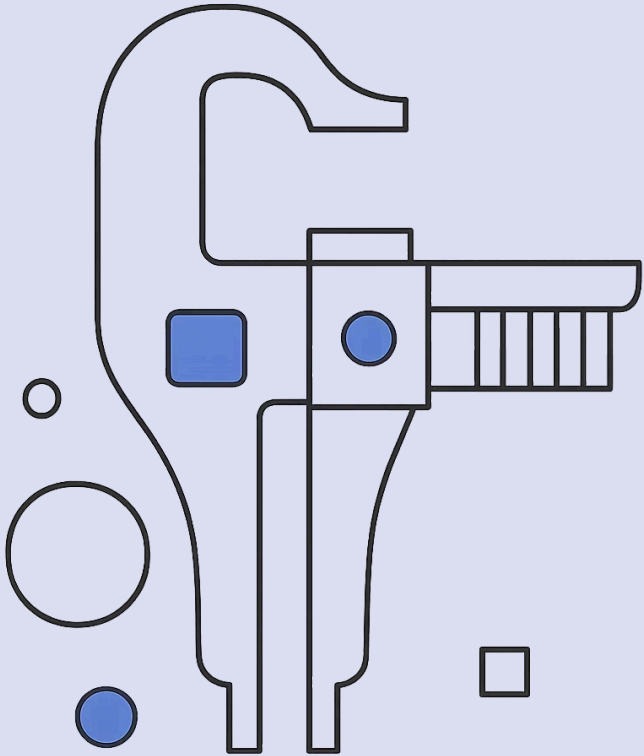


The Problem

Highly related features can make coefficients unstable.

Examples

Feature 1	Feature 2
Study Hours	Practice Tests
Attendance	Class Participation



Probability Calibration

If a model says 80% probability, ideally 8 out of 10 similar cases should be positive.

Calibration checks whether predicted probabilities are reliable and trustworthy.

Multiclass Logistic Regression

Logistic Regression can be extended to multiple classes.

Example

Grade prediction: A / B / C / D

One-vs-Rest

Train a separate classifier for each class against all others

Multinomial

Model all classes simultaneously

Strengths of Logistic Regression

01
10

Binary Classification

Probability output included



Interpretable

Coefficients are explainable



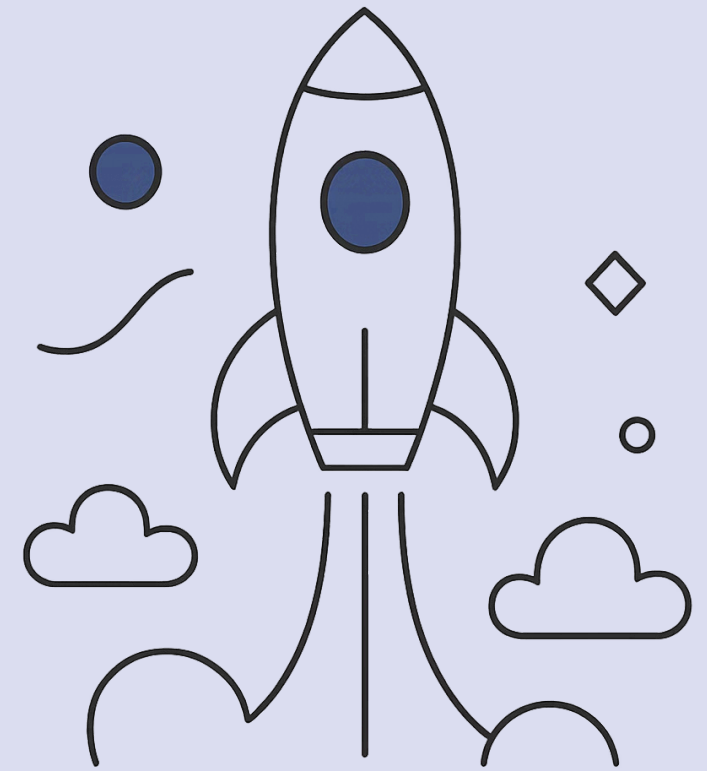
Fast Training

Strong baseline classifier



Linear Patterns

Works well with clean, separable data



Limitations

Linear Boundary Only

Needs feature engineering for nonlinear patterns

Sensitive to Outliers

Needs scaling for best optimisation

Complex Interactions

Can struggle without manual feature crafting

Imbalanced Data

Accuracy can mislead — use precision, recall, or F1